File permissions in Linux

Project description

Permissions for the research team have not been updated to match the required authorization. I have analyzed and corrected these permissions within the `projects` directory to improve the security posture of the organization.

Check file and directory details

As seen in the screenshot below, I used linux commands to view and analyze the existing permission set for `projects` directory.

```
researcher2@6f5fdb43c127:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Feb 19 23:23 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Feb 19 23:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Feb 19 23:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_t.txt
researcher2@6f5fdb43c127:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb 19 23:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb 20 00:10 ..
-rw--w---- 1 researcher2 research_team   46 Feb 19 23:23 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Feb 19 23:23 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Feb 19 23:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Feb 19 23:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_t.txt
```

The first command I used was the `ls` command to check the permissions of the directory and the files within it. Then I used the `ls` command with the `-la` option to include the hidden files in the analysis. This directory includes one hidden file named `.project_x.txt`, five other files and one directory named `drafts`.

Describe the permissions string

The 10-character string in the first column of the file contents is the permissions string. This string details the file or directory permissions for each owner type. Below I have detailed each section of the string.

- First character: If it is a `d`, its a directory and if its a hyphen (`-`), its a regular file.

- 2nd-4th characters: These characters represent the read (`r`), write (`w`) and execute (`x`) permissions for the user. If it's a hyphen (`-`) then it does not have that permission.
- 5th-7th characters: These characters represent the read (`r`), write (`w`) and execute (`x`) permissions for the group. If it's a hyphen (`-`) then it does not have that permission.
- 8th-10th characters: These characters represent the read (`r`), write (`w`) and execute (`x`) permissions for all the other users. If it's a hyphen (`-`) then it does not have that permission.

Change file permissions

The organization does not want other to have write access on any files so I found that the file `project_k.txt` had incorrect permissions. The following code shows what I did using Linux commands:

```
researcher2@6f5fdb43c127:~/projects$ chmod o-w project_k.txt
researcher2@6f5fdb43c127:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb 19 23:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb 20 00:10 ..
-rw--w---- 1 researcher2 research_team   46 Feb 19 23:23 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Feb 19 23:23 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Feb 19 23:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_t.txt
researcher2@6f5fdb43c127:~/projects$ chmod g-rw project_m.txt
researcher2@6f5fdb43c127:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb 19 23:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb 20 00:10 ..
-rw--w---- 1 researcher2 research_team   46 Feb 19 23:23 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Feb 19 23:23 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_k.txt
-rw------- 1 researcher2 research_team   46 Feb 19 23:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_t.txt
```

I used the `chmod` command to correct these permissions and remove the write permission from the other for the `project_k.txt` file. I also corrected the `project_m.txt` file since it is a restricted file and should not be readable or writable by the group or other.

Change file permissions on a hidden file

I have been informed that the `.project_x.txt` file is an archived file that should not be written to by anyone but the user and group should still be able to read it.

```
researcher2@6f5fdb43c127:~/projects$ chmod u-w+r,g-w+r .project_x.txt
researcher2@6f5fdb43c127:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb 19 23:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb 20 00:10 ..
-r--r----- 1 researcher2 research_team   46 Feb 19 23:23 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Feb 19 23:23 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_k.txt
-rw------- 1 researcher2 research_team   46 Feb 19 23:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_t.txt
```

As you can see, the `.project_x.txt` file is hidden since the name starts with a period (.). Using `chmod`, I adjusted the permissions to make sure that the only permissions this file has is for the user and group to be able to read it.

Change directory permissions

The final change my organization wanted to fix is the permissions of the drafts directory. Only the user should be able to access and execute the directory. I did the following in linux to achieve this:

```
researcher2@6f5fdb43c127:~/projects$ chmod g-x drafts/
researcher2@6f5fdb43c127:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb 19 23:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb 20 00:10 ..
-r--r----- 1 researcher2 research_team   46 Feb 19 23:23 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Feb 19 23:23 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_k.txt
-rw------- 1 researcher2 research_team   46 Feb 19 23:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Feb 19 23:23 project_t.txt
```

I removed the execute permission from the group so that now only the user has full permissions and the group and others have none.

Summary

Linux commands such as `ls -la` helped me analyze and `chmod` helped me change the permissions of hidden and viewable files as well as directories. My organization instructed me on the permission changes and I was able to put these changes into place to better the security posture of the organization.